

Deep learning-based detection of dislocation defects in 4H-SiC substrates via enhanced photoluminescence imaging[☆]

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ABSTRACT

This paper introduces a methodology for the detection of dislocation defects in silicon carbide (SiC) substrates through the application of photoluminescence (PL) imaging technology in conjunction with the you only look once (YOLO) object detection algorithm. Dislocation defects in 4H-SiC substrates have a considerable influence on the device performance, thereby necessitating accurate and rapid detection methods. However, the identification of these defects in PL images is often complicated by their low contrast and indistinct boundaries. In this study, image enhancement methods were applied to improve the quality of PL images, while the YOLO11-ORB (Oriented Bounding Box) model was used to effectively detect and localize basal plane dislocations (BPDs) in 4H-SiC substrates. The experimental results demonstrated notable improvements in defect detection and localization. Additionally, the use of the class-balanced multi-sampling (CBMS) technique enhances the distribution of classes within the dataset, thereby increasing the model's ability to identify minority defects such as BPDs. This study offers an innovative strategy for the efficient nondestructive detection of dislocation defects in 4H-SiC substrates and highlights the advantages of combining deep learning techniques with image enhancement methods.

1. Introduction

Silicon carbide (SiC) material has excellent performance, in high-power electronic devices, radio frequency uses, and high-temperature Settings are widely used in these aspects. In the field of power electronics and the emerging energy industry, the demand for SiC substrates has increased greatly, but dislocation defects exist in the SiC substrate, that is, the base surface dislocation. These defects have a negative impact on the function of the device and may even cause the breakdown of the device [1,2]. Therefore, it is critical to detect and characterize these defects in a timely and accurate manner during the production of SiC substrates.

Traditional dislocation detection techniques, such as transmission electron microscopy (TEM), X-ray diffraction (XRD), and other similar methods [3–5], are typically destructive and are not suitable for in-line detection in manufacturing processes. To address this issue, this study primarily concentrates on non-invasive testing techniques, with a particular emphasis on photoluminescence (PL) technology, which stands out because of its high sensitivity, nondestructive nature, and

superior spatial resolution [6,7]. The PL method involves exciting the 4H-SiC substrate to generate spectral signals, which results in variations in the brightness or tone of the PL images, indicative of dislocation defects. By observing these variations, much key data can be obtained, which helps detect defects in 4H-SiC materials [8]. Nonetheless, the high doping density of the 4H-SiC substrate often presents challenges in PL imaging, such as the low contrast of defects and indistinct boundaries [9]. These issues complicate the accurate localization and characterization of defects. Therefore, there is a pressing need to develop novel techniques that can effectively detect defects while preserving the integrity of the images produced.

In recent years, significant advancements have been achieved in deep learning. Specifically, in the realm of object detection technology, these advancements have led to the development of novel approaches for automatic defect detection. The you only look once (YOLO) algorithm, known for its robust real-time performance and precise localization capabilities, has gained widespread application in object detection. This effectively facilitates the target localization and classification. Compared to conventional object detection techniques, YOLO

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models demonstrate superior speed, accuracy, and flexibility, rendering them particularly adept at analyzing intricate defect patterns and extensive image datasets [10–13].

Although the YOLO model has been widely used in the detection of metal surface defects [14–16], its effectiveness in analyzing PL images of 4H-SiC has yet to be established, which presents a distinct challenge compared to metal surfaces. The 4H-SiC substrates face specific obstacles in PL imaging, such as low image contrast due to high doping concentrations [9]. Furthermore, the spectral variations associated with dislocation defects are often subtle, compounded by background noise and irregular brightness fluctuations, complicating the detection process and potentially undermining the model's reliability [17–19]. Therefore, applying YOLO to 4H-SiC PL images requires addressing these challenges and customizing the model accordingly.

To overcome these limitations, this study presents a novel approach for identifying dislocation defects in 4H-SiC substrates. The proposed method leverages PL imaging technology in conjunction with the YOLO target-detection algorithm. By employing advanced image enhancement techniques, this method enhances the quality of the PL images and integrates the YOLO11 target detection algorithm, which can accurately identify and locate basal plane dislocations (BPDs) within the 4H-SiC substrate. Notably, the YOLO11 model incorporates a directional bounding box feature, allowing it to effectively manage defects characterized by rotation and irregular shapes, thereby significantly enhancing detection accuracy. In addition, this study uses data pre-processing techniques, namely class-balanced multi-sampling (CBMS) technology, to alleviate class imbalance issues in the dataset before performing object detection, with the aim of improving the stability and overall performance of the detection model.

The research methodology described in this paper proceeds as follows. Initially, PL images of a 4H-SiC substrate were captured using a self-designed and built PL Imager System. Subsequently, the image was preprocessed using a series of image enhancement techniques, including Gaussian speckle homogenization, brightness and contrast adjustment, and denoising. The purpose of these preprocessing steps is to improve the visibility of defects and detection accuracy. Ultimately, the YOLO11-OB model was employed in conjunction with CBMS for target detection, enabling the precise identification of dislocation defects and their respective locations. This approach demonstrates the potential of integrating image enhancement methods with deep learning-based object detection techniques to effectively identify dislocation defects in 4H-SiC substrates, thereby validating the efficacy of the proposed method.

2. Experiment

A series of experiments was conducted to assess the efficacy of the dislocation defect detection method for 4H-SiC substrates, which employs photoluminescence imaging in conjunction with the YOLO11-OB target detection algorithm. The following sections provide a summary of the implementation procedures and evaluation criteria used in the experiments.

2.1. PL imaging

2.1.1. Sample preparation and excitation sources

The samples were prepared from 6-inch commercial N-type 4H-SiC substrate wafers (4° off-cut, Chemical Mechanical Polishing (CMP) polished, $R_a \leq 0.2 \text{ nm}$, specific thickness of $350 \mu\text{m} \pm 15 \mu\text{m}$, and threaded dislocation density $\leq 500 \text{ cm}^{-2}$) obtained from three manufacturers: Beijing TankeBlue, Shanghai SICC, and Suzhou Jingling. To assess the potential of utilizing PL technology in conjunction with the YOLO algorithm for detecting dislocation defects on 4H-SiC substrates, wafers from various manufacturers were sectioned into $4 \text{ mm} \times 4 \text{ mm}$ square samples of specific dimensions for experimentation. Before testing, all samples were cleaned as described in the literature [20].

A 25 mW He–Cd laser (operating at a wavelength of $\lambda = 325 \text{ nm}$) was employed for excitation, with the emission line having a penetration depth of approximately $7 \mu\text{m}$ [21–23]. The laser beam, incident on the sample surface at an angle of approximately 40° [24], exhibits an elliptical shape. The illuminated area was measured to be approximately 1.1 mm^2 , with a corresponding power density of approximately 520 mW/cm^2 at the focal spot.

2.1.2. PL image acquisition

The setup of the imaging system is illustrated in Fig. 1. The specimen was positioned on a motorized X-Y-Z displacement stage with a high precision (positioning accuracy $\leq 2 \mu\text{m}$). Imaging was performed using a stepwise fixed-point photography approach with each step involving a displacement of $50 \mu\text{m}$ in a single direction. The imaging system was positioned perpendicular to the displacement stage and a high-sensitivity industrial camera was used for image acquisition. The camera featured a resolution of $4096 \times 3000 \text{ pixels}$ and a pixel size of $3.45 \mu\text{m} \times 3.45 \mu\text{m}$, facilitating the effective capture of intricate surface details of the sample. Moreover, the camera exhibited sensitivity to infrared light with wavelengths of $\geq 700 \text{ nm}$, rendering it suitable for the infrared imaging requirements of the experiment.

The camera utilizes dual telecentric lenses and a long-wavelength pass filter ($\geq 700 \text{ nm}$) to exclusively permit the transmission of infrared light. The optical setup incorporates LBTEK optical elements. A light shield encloses the imaging optical pathway to reduce external light disturbances and enable precise imaging. A 20X magnification infrared objective lens was affixed to the system's base to improve the image resolution. Imaging was performed at ambient temperature with a 1-second exposure duration to capture high-fidelity image data from a specified area ($2.048 \times 1.5 \text{ mm}^2$).

The experimental computing platform utilized a workstation equipped with an NVIDIA RTX 4090 GPU, which offered ample computational capacity for analyzing extensive PL image datasets. Training was performed using the PyTorch deep-learning framework on Windows OS.

2.1.3. PL image enhancement dataset preparation

As shown in Fig. 6(a), the high doping concentration of the 4H-SiC substrate resulted in elevated background brightness, which diminished the contrast of the PL signal and obscured the defect regions [7,9,25]. To address this challenge, image preprocessing and enhancement were performed to improve the visibility of the defect regions, as illustrated in Fig. 6(b). Through the integration of PL imaging and target detection methodologies, the automated localization and classification of dislocation defects were accomplished.

To accurately calibrate the dislocation defects, PL imaging was augmented with destructive etching using a molten eutectic KOH-NaOH solution at 500°C for 40 min. A specific etching protocol can be found in literature [20]. Owing to the restricted penetration depth of UV-excited light, the induced defects are predominantly situated close to the wafer surface [26]. A comparison of the etched image (Fig. 6(c)) with the original image allowed for precise calibration of dislocation defects, facilitating the construction of the PL image enhancement dataset.

2.2. Image enhancement

The original PL images exhibited uneven illumination, low contrast, prominent halo effects, and background noise interference. The Gaussian distribution of the laser spot results in a bright center and darker edges, potentially obscuring weak defect signals and compromising detection accuracy. Additionally, the low contrast hinders a detailed distinction, whereas the halo effect can lead to erroneous defect identifications. Furthermore, the background of PL images often lacks pertinent information and may introduce noise, diminishing the reliability of the analysis. Therefore, enhancing the image by balancing illumination, improving contrast, mitigating halo effects, and

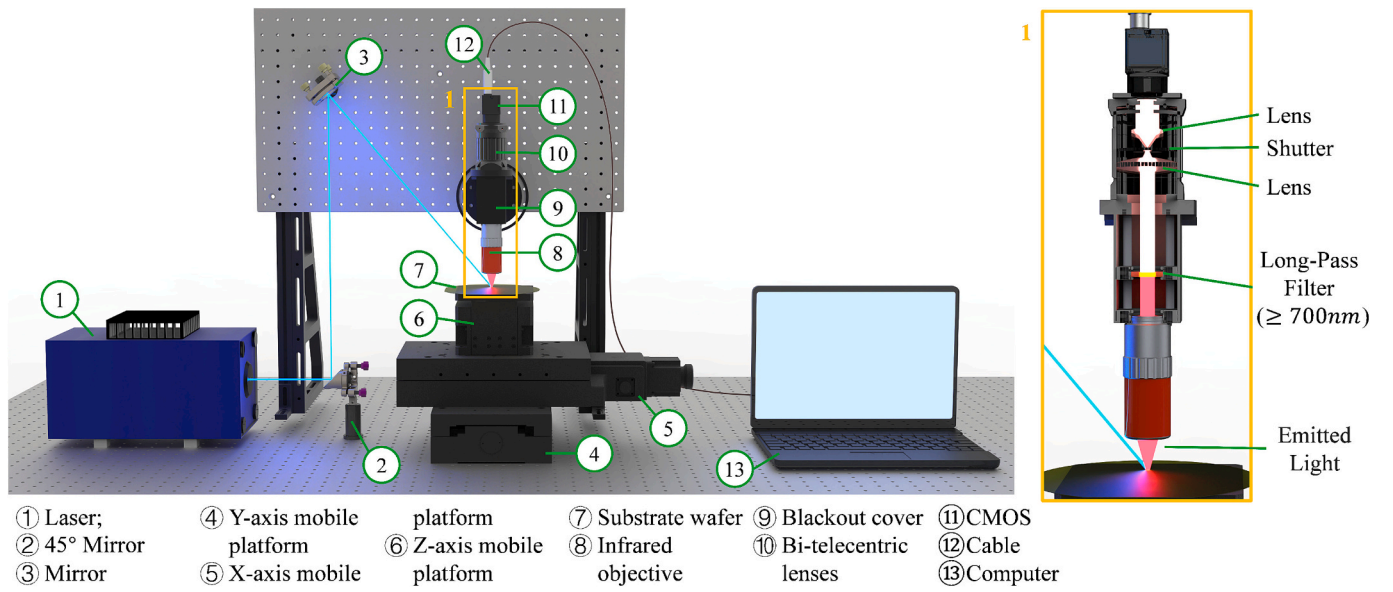


Fig. 1. PL imager system schematic.

emphasizing regions of interest is crucial for enhancing PL image usability and ensuring the precision and consistency of subsequent target detection.

2.2.1. Image enhancement steps

Gaussian Spot Extraction and Adaptive Blurring: The median stacking method was used to construct a “Gaussian spot” template, which was smoothed using a 71×71 Gaussian kernel to remove noise and emphasize the target features [27,28].

Flat Field Correction and Weighted Blending: The image was subjected to flat field correction using the “Gaussian spot” template to eliminate substrate or light source inhomogeneity. The weighted blending of the original and corrected images preserved the details while clarifying the defect regions.

Contrast and Brightness Enhancement: Contrast enhancement was achieved through the application of contrast-limited adaptive histogram equalization (CLAHE), while linear transformations were employed to adjust the image’s brightness and contrast, thereby highlighting the details of defects more effectively [29,30].

Noise reduction: Noise reduction is applied to the background area while enhancing the clarity of the target region with the aim of strengthening the contrast of the target region. The denoising of the PL images in this process was achieved using the “Topaz Photo Ai” software [31].

Gaussian Spot Center Intelligent Cropping: The image is cropped based on the centroid of the Gaussian spot “Gaussian spot,” automatically focusing on the area near the “Gaussian spot” in the image and trimming unnecessary dark edges.

Image Batch Processing: A batch process was implemented to automatically process images in a folder and save the results to a designated output folder.

2.2.2. Image quality assessment (IQA)

After AI denoising, some PL images may appear blurry, making it difficult to rely solely on the grayscale histogram for assessing the enhancement effect. Quantifying the quality of the enhanced images is crucial to ensure that they meet the necessary standards and provide reliable input for subsequent YOLO object detection. Multidimensional quality metrics were employed, focusing on defect area prominence and detail enhancement. Emphasis is placed on prioritizing defect visibility over noise suppression during optimization. In addition to subjective

evaluation, five key metrics were selected across three dimensions to objectively evaluate the enhancement effect: contrast between defect areas and background, defect detail enhancement, and visual saliency [32,33]. These metrics facilitate a quality comparison between pre- and post-enhancement images, guaranteeing that the enhanced images satisfy the quality criteria for subsequent YOLO object detection.

Dark Region Contrast (DRC): The average luminance contrast between the dark region and the surrounding background in the image (formula (1)). A higher contrast indicates a clear distinction between the dark region and the background, aiding in better defect identification.

$$DRC = |\mu_{\text{dark}} - \mu_{\text{background}}| \quad (1)$$

where μ_{dark} is the average luminance of the dark region, and $\mu_{\text{background}}$ is the average luminance of the background region.

Dark Detail Entropy (DDE): This metric evaluates the level of detailed information in the background region surrounding the dark spots by calculating the grayscale distribution of the background area, thus measuring the complexity of the image background (formula (2)). A higher entropy value indicates a more detailed background information.

$$DDE = - \sum_i P_{\text{dark}}(i) \log P_{\text{dark}}(i) \quad (2)$$

where $P_{\text{dark}}(i)$ is the grayscale level probability of the dark region, which can be obtained by calculating the frequency of the occurrence of each grayscale level within the dark region.

Visual Saliency- ITTI algorithm (VS-ITTI): Itti et al. proposed a visual attention model in 1998 that integrates image color, brightness, and orientation features using a Gaussian pyramid, which exhibits good robustness against noise [34]. The simplified ITTI algorithm used in this study utilizes only brightness features [35,36]. In this case, visual saliency is based on the brightness channel (L channel) of the image, and saliency is measured by calculating the standard deviation of the brightness map. The process involves first converting the image to the Lab color space, extracting the L channel, and then calculating its standard deviation as a measure of saliency, which can be represented by formula (3):

$$VS = \sigma_{L-channel} \quad (3)$$

where $\sigma_{L-channel}$ is the standard deviation of the L-channel (luminance).

Visual Saliency-Context-Aware (VS-CA): The CA algorithm is a saliency detection method proposed by Goferman et al. in 2010 [37] that is based on both local and global features. The authors argue that a saliency map should include both the target and the surrounding background area to better represent the saliency information of the image. The specific calculation formula is as formula (4):

$$CA = \sigma \left(\sum_{ij} w(i,j) I(x+i, y+j) \right) \quad (4)$$

where $w(i,j)$ is the Gaussian kernel, $I(x,y)$ is the pixel value of the original image, σ is the standard deviation of the Gaussian kernel, and i and j are the offsets relative to the center pixel.

Modified No-Reference Peak Signal-to-Noise Ratio (MNRPSNR): Image quality evaluation method based on visual masking effects and gradient information. By automatically adjusting the image block segmentation and noise detection thresholds, MNRPSNR can avoid the issue of manually setting fixed thresholds in traditional methods, providing a quality assessment that is more adaptable to different types of noise. In addition, the algorithm enhances the consistency of the evaluation results with human subjective perception through correction and normalization processing [38,39].

2.2.3. Image quality metrics (IQM) program

To ensure precise evaluation results by focusing on key areas and minimizing background noise interference, all IQMs were computed using image data within the designated IQM calculation region. Given the distinct manifestations of threading dislocations (TDs) and BPDs in PL images (TDs as dark spots and BPDs as dark line structures) evaluation algorithms have been tailored for dark spot and dark line images to account for these specific defect characteristics, as illustrated in Fig. 2. The specific workflow is described below.

Both programs adopted a local regional evaluation strategy. The IQM program for dark spots first identifies a single defect (Fig. 3(a)) and

marking the locations of the dark spot with red dots (Fig. 3(b)), then extracts its defect contour from the enhanced PL image, as indicated by the green box in Fig. 3(c). Next, based on the center of the extracted defect contour, a 50×50 pixel square IQM calculation region was expanded, as shown by the red box in Fig. 3(c). Subsequently, the same contour and IQM calculation regions are marked at the same location in the original image Fig. 3(d). Finally, the calculation of each image quality evaluation metric is completed within the IQM calculation region.

The IQM program for dark lines first selects a 50×200 pixel IQM calculation region, as shown by the blue box in Fig. 4(b). The dark line is then identified within the IQM calculation region (the yellow line in Fig. 4(c)). Next, it displays and confirms the selected calculation region and dark line in the enhanced PL image (the red box and yellow line in Fig. 4(d)). Finally, the same dark line and IQM calculation region were marked on the original PL image (red box and green lines in Fig. 4(e)), and the image quality metrics for each image are calculated within the calculation region. To guarantee the representativeness and statistical reliability of the evaluation outcomes, 100 samples were gathered for each defect type. A thorough and precise assessment of image quality can be achieved by analyzing the variations in the metrics of defects located in the same area.

2.3. Object detection algorithm

2.3.1. YOLO model

Model Architecture: The YOLO model is a deep-learning object detection system that employs Convolutional Neural Networks (CNNs) to simultaneously classify and locate objects. Its main component, the head, contains three detection branches, each of which is dedicated to detecting objects of different sizes. YOLO11 offers multiple versions, such as YOLO11-Small, YOLO11-Medium, and YOLO11-Large [11,13]. For this experiment, the YOLO11-Large version was utilized, considering the captured images and the target size.

Dataset Preparation: A comparison was conducted using an etched image of the same region. Dislocation defects in the enhanced PL image were manually annotated using the open-source annotation tool “X-AnyLabeling [40].” The position of each defect was

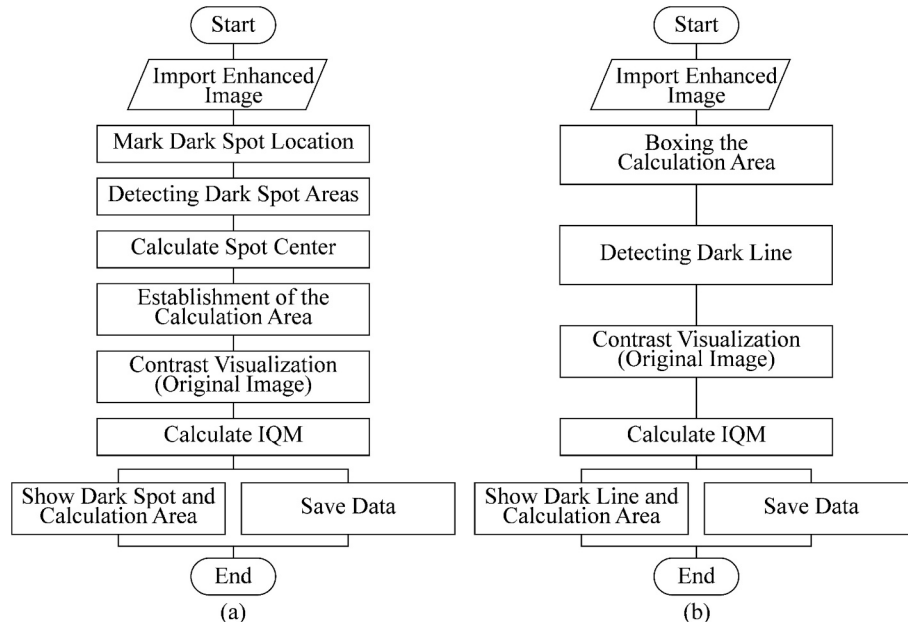


Fig. 2. Flowcharts for IQM calculation procedures for (a) dark spot and (b) dark line image evaluation.

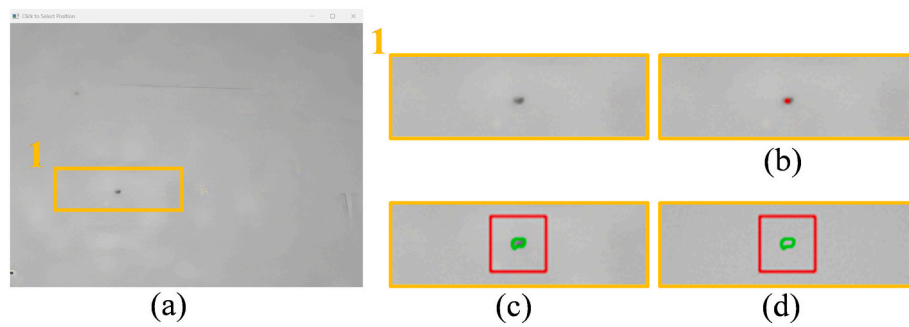


Fig. 3. Schematic diagram of the region for dark spot image evaluation program: (a) Input enhanced PL image. (b) Manual calibration of dark spot location (red dot). (c) Automatically extracted dark spot contours (green box) and IQM calculation area (red box). (d) Display of dark spot contours (green box) and IQM calculation area (red box) at the same location in the original PL image.

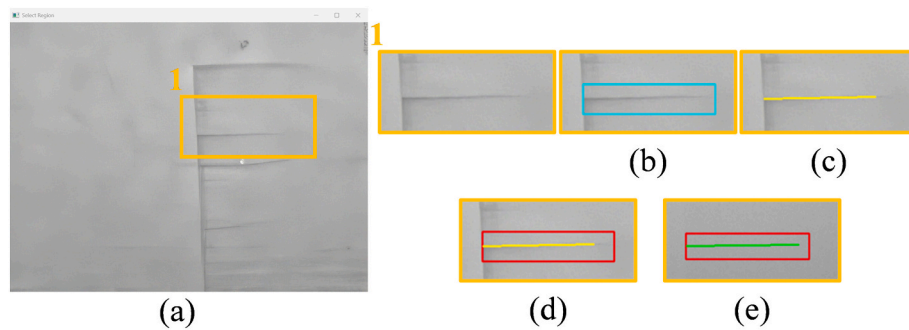


Fig. 4. Schematic diagram of the region for dark line image evaluation program: (a) Input enhanced PL image. (b) Manual selection of the calculation area (blue box). (c) Automatically extracted dark line contours (yellow line). (d) Display of dark line (yellow line) and IQM calculation area (red box) in the enhanced PL image. (e) Display of dark line (green line) and IQM calculation area (red box) at the same location in the original PL image.

documented, including the center coordinates, width, height, rotation angle, and defect categories. The TDs are denoted by green labels, whereas the BPDs are denoted by brown labels, as illustrated in Fig. 5. Our self-built dataset is divided into two subsets, including a training set with 2092 images and a validation set with 489 images. **Training Process:** Training was performed using the open-source YOLO11-OB framework, adjusted for the experimental data. Each training cycle, referred to as an epoch, comprises several batches, and early stopping techniques are implemented to mitigate the risk

of overfitting. If the model's mean average precision (mAP) did not improve over a series of consecutive calendar epochs, the training process was terminated early.

Hyperparameter Settings: A Stochastic Gradient Descent (SGD) optimizer was employed, integrating classification loss, target box loss, and intersection over union (IoU) loss to effectively balance the optimization of category detection alongside the parameters of the bounding box, which encompass the position and rotation angle. According to the training results of preliminary experiments, the

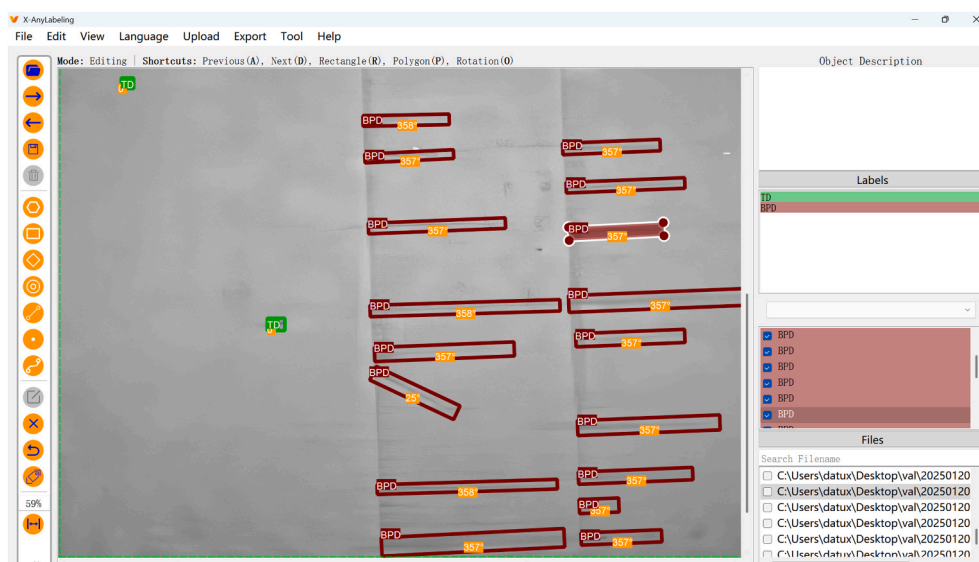


Fig. 5. Schematic of dislocation defect annotation in PL images.

model's indicators tend to converge around 300 epochs. Therefore, in the formal training, the learning rate was initially established at 0.01, with a batch size of 8 and the number of training cycles is set to the maximum training cycle of 300 in the preliminary experiment.

Model Optimization: The CBMS strategy was adopted to improve the detection accuracy for all classes. Samples from underrepresented classes were replicated to increase their frequency in the training set, whereas samples from overrepresented classes were reduced to ensure a balanced class distribution.

2.3.2. Evaluation metrics

Precision: This refers to the ratio of true positive samples to the total number of samples predicted as positive by the model.

Recall: This quantifies the ratio of true-positive samples that the model accurately identifies.

mAP: The overall metric was derived from the mean of the Average Precision (AP) values for all categories. The formula for AP is as formula (5):

$$AP = \int_0^1 Precision(r) dr \quad (5)$$

Thus, formula (6) for mAP is:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (6)$$

In object detection evaluations, the mAP serves as a fundamental metric. It encompasses the precision of the model's bounding box localization, quantified by IoU, as well as the classification accuracy.

- **mAP50 (Mean Average Precision at IoU 0.5):** This represents the mean precision at an IoU threshold of 0.5.
- **mAP50-95:** This represents the mean precision calculated over IoU values that span from 0.5 to 0.95, utilizing a step increment of 0.05.

3. Results and discussion

3.1. Comparison and analysis of image enhancement effects

3.1.1. Dislocation defect resolution and localization in PL images

Previous research has demonstrated that dislocation defects in PL images appear as circular dark spots known as TDs and dark lines of varying lengths referred to as BPDs [25]. Owing to the high doping concentration of the substrate, enhancing the PL images is often necessary for clear visualization of dislocation defects. As shown in Fig. 6, the enhanced images significantly improve the recognition of dislocation defects. The original PL images (Fig. 6(a)) show an uneven light distribution due to Gaussian spots, which obscure the defect regions. After the image enhancement, the Gaussian spots were homogenized, resulting in

a uniform light distribution and more prominent dislocation defect regions (Fig. 6(b)). Further denoising significantly reduces the noise and enhances the prominence of the dislocation regions (Fig. 6(c)). Finally, the etching image (Fig. 6(d)) was used for further classification and localization of the dislocation defects in the PL images.

Through defect classification and localization analysis of the PL image, it was found that the PL image containing BPD (purple box) shows dark lines (yellow dashed box) similar in shape to BPD; however, no corresponding BPD etch pits can be found in the etched image. According to previous research, dislocation defects are considered to be growth defects [41], and their depth in the wafer can influence the results of both PL and etch detection. It is speculated that the mismatch between the PL image and the etch results may be due to differences in the etch depth and PL penetration depth. To verify this hypothesis, the wafer thickness was measured before and after etching. The results showed that under the current etching parameters, the etch depth was approximately $3 - 5 \mu m$, compared to a laser penetration depth of $7 \mu m$ [21], allowing it to capture more dislocation information. In particular, after Chemical Mechanical Polishing (CMP) processing, BPD tends to accumulate more easily in the surface scratch areas, leading to the appearance of multiple dark lines in the PL image. However, these dark lines may not correspond to the etch pits in the etched images. Based on this, we believe that these dark lines should be classified as BPD and will continue to be labeled BPD in subsequent object detection tasks.

3.1.2. Quantitative evaluation of image enhancement effects

The grayscale histograms of the PL images (Fig. 6(a)–(c)) cannot fully reflect the extent of defect region enhancement and detail improvement, especially after batch processing of the PL images through Topaz AI denoising, where the images may sometimes become blurry. Therefore, quantitative metrics were introduced to further evaluate the enhanced PL images in multiple dimensions. Based on these evaluation results, image quality standards were established for subsequent YOLO detection to filter out images that meet the quality requirements, ensuring that they can effectively support the object detection task.

As shown in the cloud rain plot in Fig. 7, the enhanced image outperformed the original image in several metrics. Overall, all metrics showed a more concentrated distribution after enhancement (blue cloud rain diagram), indicating that the image quality was more consistent and improved. Only the MNRPSNR metric (Fig. 7(a)) metric decreases after enhancement, suggesting that while the enhanced image excels in visual quality, it compromises on noise suppression.

In other metrics, DRC (Fig. 7(b)) shows that the enhanced image has a higher contrast, indicating richer brightness and color layers, with clearer details. The DDE (Fig. 7(c)) reveals that the enhanced image retains more details, with a significant increase in detail complexity and information content.

For the two visual saliency metrics, VS-CA and VS-ITTI (Fig. 7(d) and

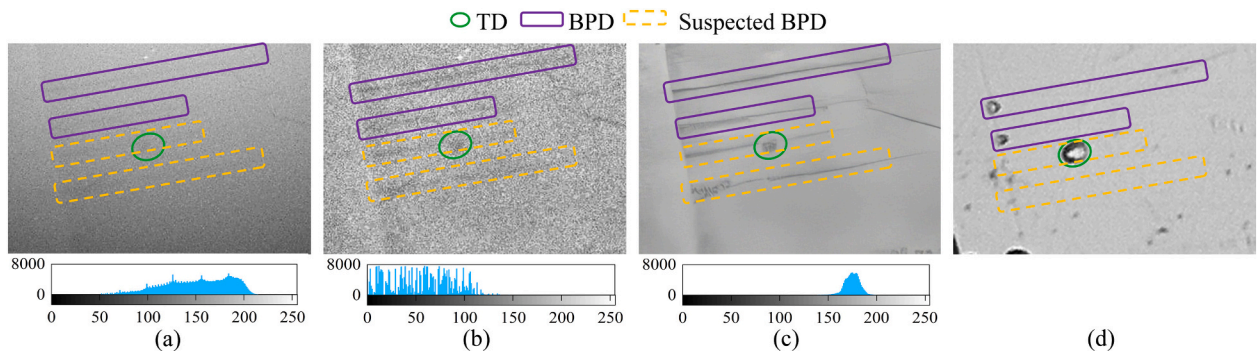


Fig. 6. (a) Original PL image and its histogram; (b) enhanced with spot homogenization and contrast adjustment and its histogram; (c) denoised PL image and its histogram for target detection input; (d) etched images from the same wafer region.

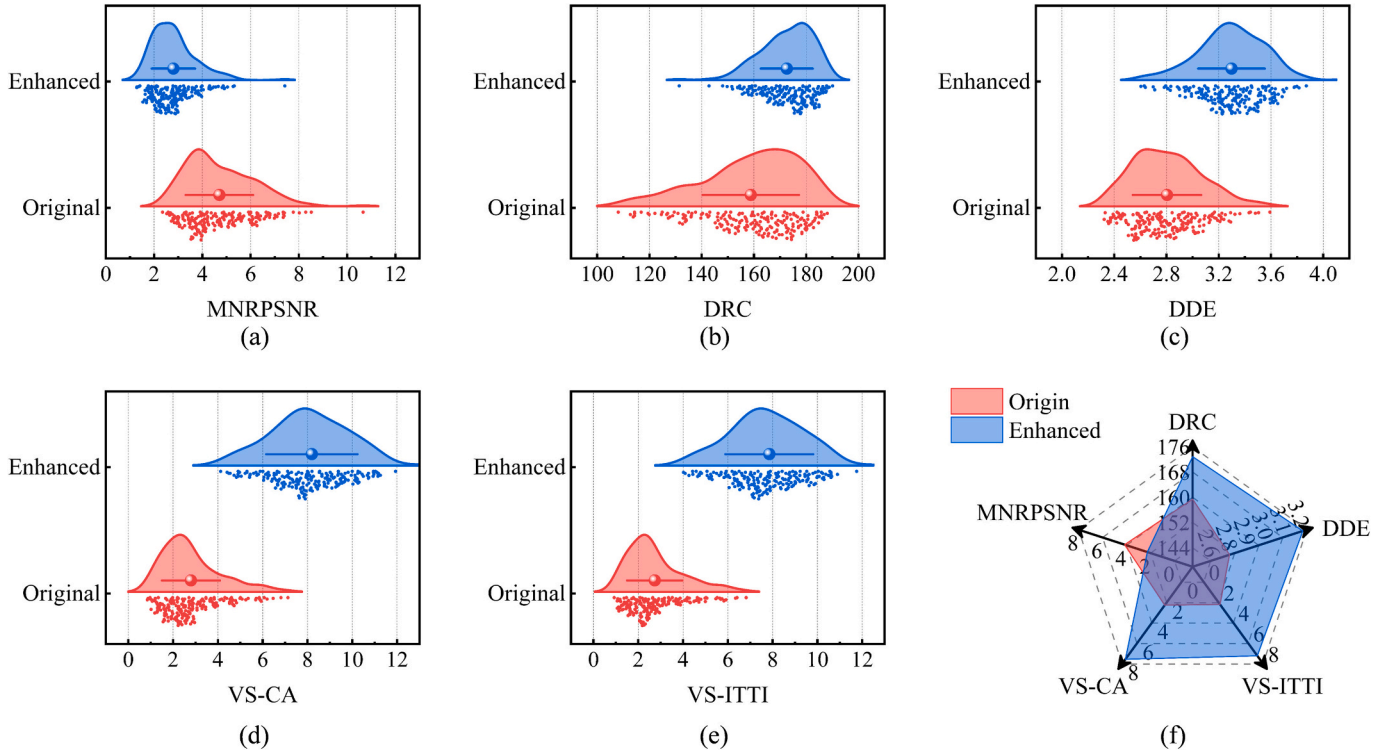


Fig. 7. Cloud rain plots of metrics before and after image enhancement: (a) MNRPSNR (b) DRC (c) DDE (d) VS-CA (e) VS-ITTI. (f) Radar chart of all metrics before and after image enhancement.

(e)), the enhanced image scores were higher in both, indicating significant improvements in contrast, detail retention, and brightness distribution. The important regions were more prominent, with enhanced brightness contrast and clearer differentiation.

The radar chart in Fig. 7(f) summarizes these results and demonstrates the superior performance of the enhanced image across multiple metrics. Despite a sacrifice in the MNRPSNR, the enhanced image clearly outperforms the original image in terms of overall quality, with advantages in contrast, detail retention, and visual saliency. The radar chart highlights the multidimensional strengths of the enhanced image, thereby confirming the effectiveness of the enhancement process.

Further evaluation of the changes in different defect types after image enhancement (Fig. 8) reveals that both dark spots and dark lines show a noticeable improvement in overall visual effect, although there are some differences in the enhancement results between the two.

In the dark-spot region, the enhancement effect is particularly significant. The enhanced image shows a higher contrast and more details, especially in terms of brightness levels and detail retention. The enhanced dark spot region appears more prominent and clearer with an increased standard deviation, indicating enhanced brightness differences and contrast, making the details more vivid.

In contrast, the enhancement effect in the dark-line region was more moderate. Although the enhancement improves the contrast and detail, the effect is not as pronounced as in the dark spot region. The brightness and details of the dark line region appear smoother, and the increase in standard deviation is smaller, suggesting that the enhancement of details is limited, and the changes in brightness contrast are relatively minor, resulting in a flatter visual effect.

To ensure that the image quality met the requirements for subsequent YOLO object detection, the image quality standard was determined using the probability density function method after statistical analysis of the sample data and comparison with actual images. Specifically, because most images have acceptable quality, only a small number of images fail to meet the standard for further object detection. Therefore, the image quality standard was set such that 99% of the images met the quality requirements, with the standard defined as the value corresponding to the 1st percentile. The image quality qualification standard is defined by Eq. (7):

$$\text{Qualification Standard} = \text{The value corresponding to the 1st percentile} \quad (7)$$

As shown in Table 1, the image must satisfy the minimum

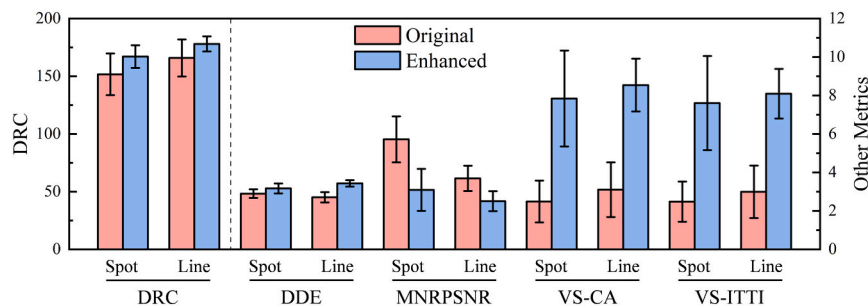


Fig. 8. Comparison of visual metrics between the enhanced image and the original image in the dark spot and dark line regions.

Table 1

PL image quality standards and minimum requirements.

Minimum requirement	DRC	DDE	MNRPSNR	VS-CA	VS-ITTI
Spot	144.24	2.56	0.55	2.04	1.92
Line	162.84	3.04	1.31	5.35	5.09

requirements for the following metrics.

When the image quality meets the above standards, it is considered to be a qualified image and can be directly used for subsequent YOLO object detection. If any metric fails to meet the standard, it indicates that the image requires further enhancement, such as improving the contrast and denoising, to ensure that the image has sufficient detail for object detection.

3.1.3. Comparison of performance with the same image enhancement

To assess the influence of image enhancement techniques on the detection efficacy of the YOLO11-OB (CBMS) model, diverse image enhancement approaches were implemented, and their efficacy in identifying 4H-SiC substrate dislocation defects was evaluated. Fig. 9 illustrates the performance of the YOLO11-OB (CBMS) model when subjected to different image preprocessing and enhancement techniques, as measured by various metrics.

As shown in Fig. 9, image enhancement plays a significant role in improving the detection accuracy of the YOLO11-OB (CBMS) model. Unenhanced PL images cannot detect defects, resulting in ineffective object detection. After the contrast and brightness adjustments, the defect signals became visible and were successfully detected. However, when relying solely on contrast and brightness enhancement, although there is some improvement in the early stages of training, the growth in precision and recall is relatively slow (Fig. 9(a)–(b)), and there is no significant improvement at high IoU thresholds. This suggests that the contrast and brightness enhancement alone are insufficient to fully improve the detection performance of the model. In particular, in the presence of noise or complex data, the performance of a model can become unstable.

In contrast, another model combines contrast improvement, brightness improvement, and noise reduction. As the training progressed step by step, the model showed a steady improvement in accuracy, recall, mAP50, and mAP50-90. In the later stages of training, the blue curve exceeds the red curve. This shows that in several evaluation indicators, the model has been more consistently improved, and after the addition of noise removal, the elasticity of the model is greatly improved. This also highlights the good effect of noise removal on the model performance, which means that noise removal can help the model improve the ability to identify objects, but also effectively reduce the negative impact of noise on the model. In complex cases, it is very meaningful to apply

denoising technology because it can effectively improve the accuracy and stability of the model in target-detection tasks.

3.2. Comparison and analysis of object detection methods

3.2.1. Comparison of the impact of bounding boxes on performance

The original version of YOLO11 uses axis-aligned bounding boxes (AABB). In this case, the bounding box surrounding the target was aligned with the coordinate axis. However, this axis-aligned bounding box is unsuitable for rotating or irregular objects. As shown in Fig. 10(a), when the target is tilted at a large angle, the detection region contains excessive background, making target recognition and classification difficult and reducing target recognition accuracy. In contrast, the oriented bounding box (OBB), which is more sensitive to direction, is more effective when dealing with rotating objects or those that are not aligned with the axis, and can provide more accurate target positioning and detection results [42]. In this work, the PL signal of BPD is linear, and the image contains objects with varying rotation angles. Therefore, we replaced the AABB in YOLO11 with OBB, as shown in Fig. 10(b). This allows BPD to be more accurately outlined while reducing background in the detection area and lowering the probability of bounding box overlap.

To evaluate the performance differences between YOLO11 and YOLO11-OB in object-detection tasks, a comparative experiment was conducted, and the results are presented in Fig. 11.

From Fig. 11, it can be observed that YOLO11 and YOLO11-OB exhibit different performance. In terms of the trends in precision and recall, the YOLO11-OB model (blue curve) showed a similar trend to YOLO11 (red curve) during most of the training process, but its precision gradually increased after approximately 50 epochs. However, it is still not possible to directly determine the impact of OBB on performance from these charts, as these individual metrics alone do not reflect the overall effect. Therefore, the precision–recall (PR) curve can serve as an important basis for further evaluating the strengths and weaknesses of the two models, as shown in Fig. 12 below.

From the PR curve (Fig. 12), it can be seen that BPDs maintain higher precision at higher recall rates, indicating that OBB plays a crucial role in detecting elongated dark-line defects (BPDs). This is because elongated dark line defects require more precise bounding box localization, and OBB provides higher localization accuracy, thereby improving the detection precision. The red curve represents the detection results for circular dark spot defects (TDs). The localization requirements for these defects are relatively low and they are less sensitive to the rotation angle of the bounding box. Consequently, the advantages of using OBB are not as pronounced, leading to lower precision at higher recall rates. Because the TDs themselves do not require precise bounding boxes, the OBB does not provide significant performance improvement.

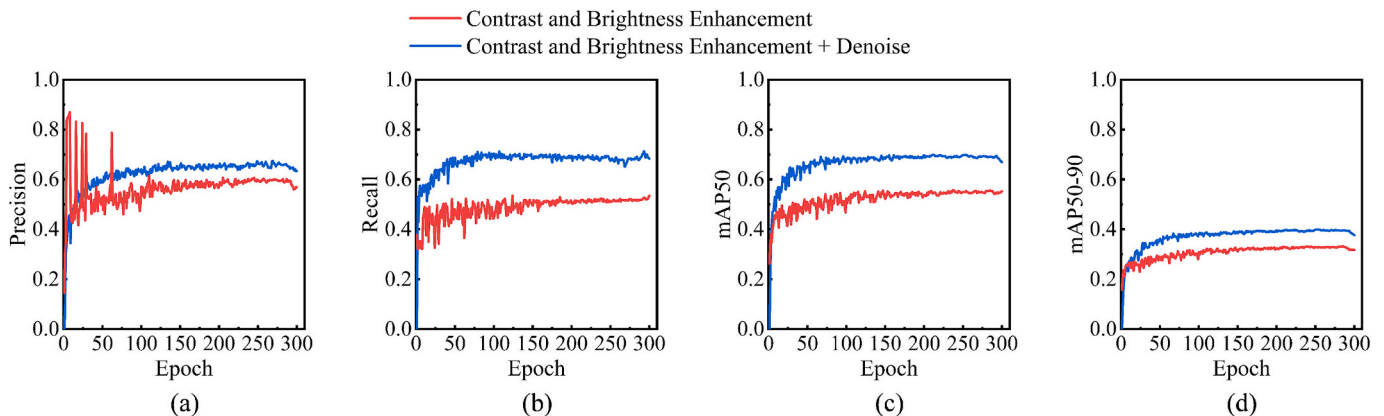


Fig. 9. The impact of the same image augmentation methods on the performance of the YOLO11-OB (CBMS) model: (a) Precision; (b) recall; (c) mAP50; (d) mAP50-90.

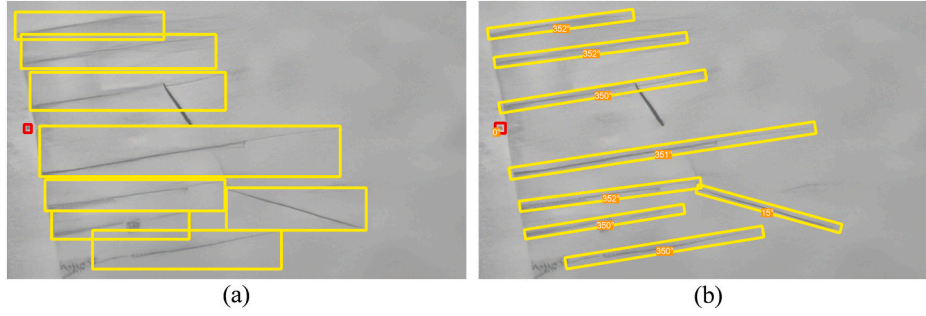


Fig. 10. Comparison of the effects of two bounding boxes on target selection: (a) using ABB (b) using OBB.

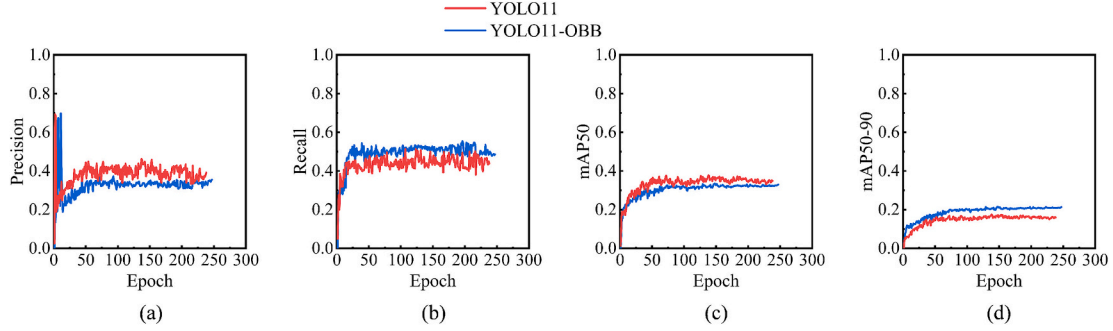


Fig. 11. Comparison of the performance between YOLO11 and YOLO11-OBB models: (a) Precision; (b) recall; (c) mAP50; (d) mAP50-90.

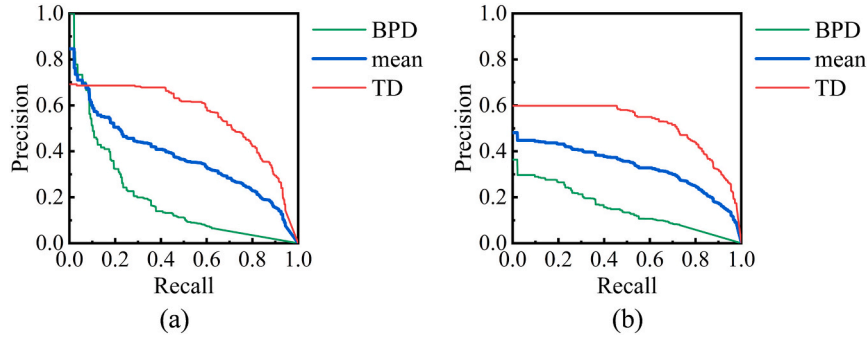


Fig. 12. Comparison of PR curves for detection performance of different defect types: (a) Without OBB; (b) with OBB.

Furthermore, in the PR curve, the blue line (mean value) in Fig. 12(b) is superiorly in the precision-recall relationship, as indicated by the blue line, when compared to the blue line in Fig. 12(a). Specifically, the PR curve with OBB (blue line in Fig. 12(b)) maintains higher precision at higher recall rates, whereas without OBB (blue line in Fig. 12(a)), the precision drops significantly as recall increases.

This indicates that using OBB can improve precision at elevated recall levels, particularly in identifying targets, such as elongated dark lines, where OBB facilitates more accurate target localization, thereby enhancing the overall detection precision. Conversely, in the absence of OBB, the detection precision is constrained by the shape and localization accuracy of the targets, resulting in a more rapid decline in precision as the recall rates increase.

3.2.2. Performance comparison of different YOLO models

Based on the previous experimental results, the OBB bounding box target detection results were significantly affected. Therefore, a YOLO model with an OBB was selected for performance comparison. YOLOv8-OBB is the earliest version in the YOLO model series to adopt OBB bounding box. In this model architecture, multi-layer convolutions are

used to extract hierarchical characteristics, and the Feature Pyramid Network (FPN) is used to improve the model's ability to detect objects of different sizes. Its algorithm structure is shown in Fig. 13(a) [11]. YOLOv10-OBB is based on YOLOv8-OBB, adopting an enhanced Cross-Stage Partial Network (CSPNet) in the main network backbone and featuring a unique Path Aggregation Network (PAN) layer. Additionally, YOLOv10-OBB introduces Non-Maximum Suppression (NMS)-Free training, utilizing a dual-label assignment strategy and a self-attention mechanism module, with the algorithm structure shown in Fig. 13(b) [11,43]. On the other hand, YOLO11-OBB builds upon YOLOv8-OBB by introducing a Cross-Stage Partial with Self-Attention (C2PSA) block. This allows the model to better capture contextual information between multiple layers, ultimately making object detection more accurate. Furthermore, in YOLO11-OBB, the C2f block has been replaced by C3k2 block. Fig. 13(c) illustrates its algorithmic structure [11,13,44].

Using the above three different algorithm models, we tested our self-built database, and the comparison results of their performance are shown in Fig. 14.

From Fig. 14, it can be seen from Fig. 12 that YOLO11-OBB (red), YOLOv8-OBB (blue), and YOLOv10-OBB (green) show similar

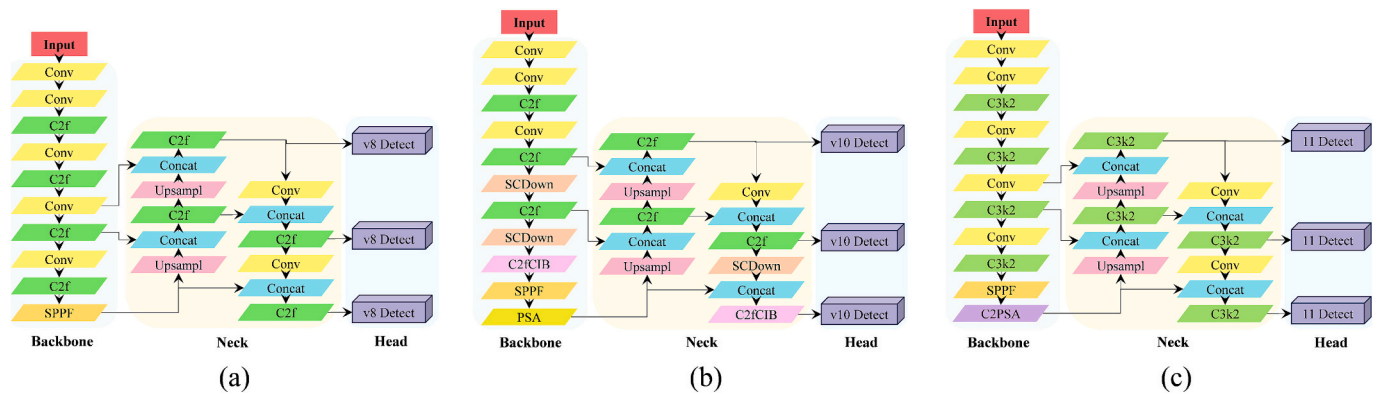


Fig. 13. Algorithm structure diagram: (a) YOLOv8-OBB, (b) YOLOv10-OBB, (c) YOLO11-OBB.

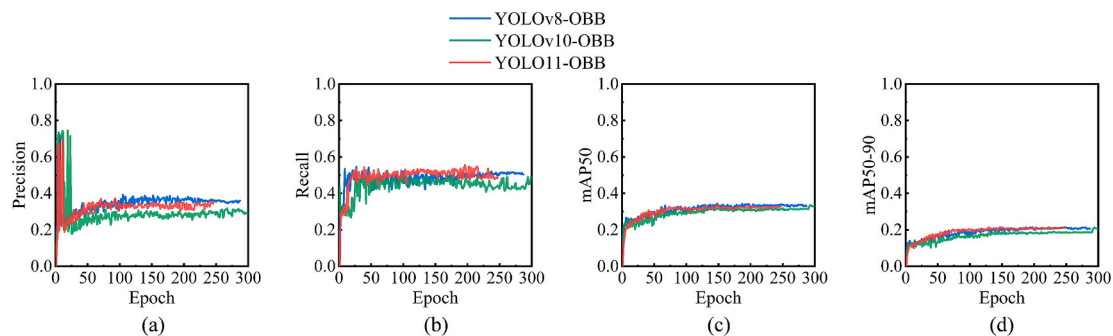


Fig. 14. Comparison of the performance of different YOLO-OBB models during the training process: (a) Precision; (b) recall; (c) mAP50; (d) mAP50-90.

performances across all metrics, but YOLO11-OBb consistently maintains a slight advantage. Although the curves for precision, recall, mAP50, and mAP50-90 for all three models are similar, YOLO11-OBb exhibits better stability with fewer fluctuations, allowing it to slightly outperform the other two models in terms of the overall performance. In the later stages, YOLOv8-OBb and YOLOv10-OBb show similar performances, especially in mAP50 and mAP50-90 (Fig. 14(c)–(d)), where the difference is minimal. Therefore, although YOLO11-OBb is slightly stronger in absolute terms, the overall difference among the three models is not very significant, suggesting that their performance in the specific task is relatively similar, with each model’s optimization strategy having a comparable impact on the results.

From an algorithmic structure perspective, YOLOv8-OBb adopts a more efficient C2f block and differentiated channel design to enhance feature extraction capabilities. The C2f block accelerates model execution through two convolutions and multiple Bottleneck blocks, achieving a faster CSPNet bottleneck layer. Its block structure diagram is

shown in Fig. 15(a). Additionally, it introduces a decoupled head and Anchor-Free architecture to enhance classification and localization performance; Task-Aligned Assigner and Distribution Focal Loss are used to optimize matching and regression accuracy; Mosaic data augmentation is disabled in the final training stage to improve training effectiveness. The overall model supports multi-resolution detection and instance segmentation, making it applicable for object recognition in complex scenarios. Therefore, YOLOv8-OBb can effectively complete the task of identifying dislocation defects on highly doped wafers [11,45].

YOLOv10-OBb and YOLO11-OBb are both new algorithms improved upon YOLOv8-OBb. Among these, YOLO11-OBb introduces the C3k2 and C2PSA blocks. The C3k2 block is an extension of the C2f block, offering the option to use the C3k block when stacking bottleneck blocks. The C3k block allows users to customize the size of convolutional blocks, providing greater flexibility. On the other hand, YOLO11-OBb adds a C2PSA block after the SPPF block. The C2PSA block is an

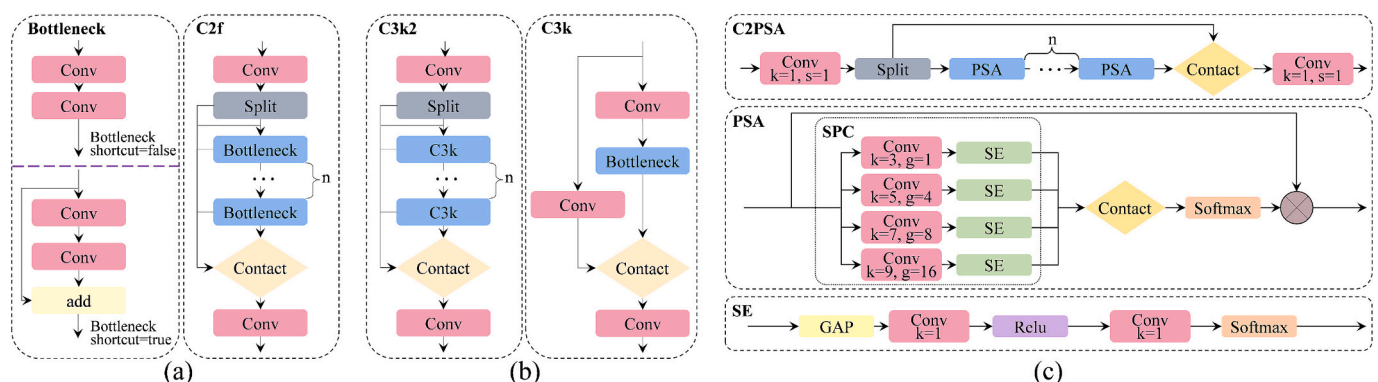


Fig. 15. Block structure diagram: (a) C2f block of YOLOv8 and (b) C3k2 block of YOLO11, (c) C2PSA block.

attention-based enhanced convolution block that applies positional attention through stacked Partial Self Attention (PSA) blocks. It splits the input tensor, applies attention to a part of it, then recombines and processes it through the final convolution. C2PSA implements a more powerful attention mechanism, enabling YOLO11-OBb to enhance its ability to capture contextual information and improve its handling of complex backgrounds and rotated objects. As a result, after prolonged training, YOLO11-OBb demonstrates more stable performance, particularly at high recall rates [13,44,45].

As for YOLOv10-OBb, despite incorporating optimizations such as CSPNet and PAN, the actual performance of YOLOv10 is not as good as expected. Since the version we used was developed by a research team from Tsinghua University, which is not an official released version, it lacks official optimizations [43]. In addition, YOLOv10 adopts a training strategy that does not require NMS, aiming to improve inference efficiency rather than inference accuracy. This may result in deficiencies in the model structure and training process, leading to a lower detection accuracy at higher IoU thresholds. This is one of the reasons why YOLOv10-OBb lagged behind the other models in all metrics.

Furthermore, all three algorithm models performed poorly in the mAP50 and mAP50-90 metrics, and there were fluctuations in their training curves, especially at higher IoU thresholds, where their performance did not meet expectations. Referring to the PR curves (Fig. 16), there are significant differences in the precision-recall curves for the two defect types (BPDs and TDs). In particular, BPDs (green) have lower overall precision, and as recall increases, precision decreases significantly, suggesting that the model may not have sufficiently learned the features of the minority class samples. On the other hand, TDs (red) performed better in certain areas but showed lower precision at lower recall rates. As recall increased, precision significantly declined, indicating that the model still suffered from large errors in detection at high recall rates.

These results suggest that the model tends to favor majority class samples when handling imbalanced data, leading to poor detection performance for minority classes. Particularly in high-recall areas, the model exhibits large fluctuations in precision, which may be due to the insufficient number of minority class samples, preventing the model from effectively learning to identify these samples during training. Therefore, to improve the overall performance, particularly in balancing precision and recall, it is necessary to adopt a class-balanced over-sampling strategy.

3.2.3. Data optimization method - class balanced multi-sampling (CBMS)

Dislocation defects in wafers are unevenly distributed, and the number of different types of dislocations varies greatly. According to our self-built dataset, the number of TDs in the image is much higher than that of BPDs. If no processing was done, the YOLO model might focus too much on the frequently occurring TDs during training and ignore the BPDs, thereby affecting the detection accuracy of BPDs. To address this

issue, we preprocess the data using the CBMS method before performing object detection. This method adjusts the sampling probabilities of each category to make the training samples more balanced, avoiding model learning imbalances caused by an excess or deficiency of samples in certain categories [46,47]. In this work, we employ random over-sampling of BPD with 1000 images to increase the frequency of this category.

Fig. 17 shows the performance improvement of the YOLOv8-OBb and YOLO11-OBb models after the introduction of the CBMS strategy, comparing the training curves of the different models.

Based on the results shown in Fig. 17, it is evident that the introduction of the CBMS strategy significantly improved the performance of the YOLOv8-OBb and YOLO11-OBb models across all metrics. In the precision, recall, mAP50, and mAP50-90 curves, models using CBMS (blue and purple lines) demonstrated more stable and higher performance than those without CBMS (red and green lines).

In the early stages of training, the YOLOv8-OBb (CBMS) and YOLO11-OBb (CBMS) models showed rapid and consistent increases in precision and recall. Their curves demonstrate elevated values and a more gradual increase, suggesting that the models are more proficient in learning from and managing data throughout the training process. Conversely, the YOLOv8-OBb and YOLO11-OBb models lacking CBMS showed more significant variations in recall and mAP metrics, accompanied by a slower rate of convergence. Moreover, YOLO11-OBb (CBMS) has always been better than YOLOv8-OBb (CBMS) for every indicator, indicating that the YOLO11 architecture receives more benefits from the CBMS strategy.

Based on the impact of algorithmic structure on performance differences in previous models, the results of this experiment indicate that data preprocessing prior to object detection plays a more prominent role in improving model performance than optimizing algorithm structure. This can be explained from several perspectives: first, deep learning models are highly dependent on data distribution. Neural networks are essentially “statistical learners” that learn patterns and distributions from training data. If the data itself has an imbalanced class distribution, the model may overfit the frequently occurring classes while neglecting rare classes, leading to performance bias. Second, object detection relies more heavily on the ratio of positive to negative samples than classification. Object detection not only classifies objects but also locates their bounding boxes. Each training sample is typically divided into “positive samples” and “a large number of negative samples” (background). If positive samples for certain categories are extremely scarce, the regressor will not learn sufficient bounding box regression signals, and the classifier will struggle to capture category features (low recall, high false negatives) [46–48]. CBMS ensures that each category has a more balanced probability of appearing during training, enabling the model to learn the discriminative features of each category more fairly, improve recall for low-frequency categories, and reduce training bias and overfitting for common categories. Among these, improving recall is

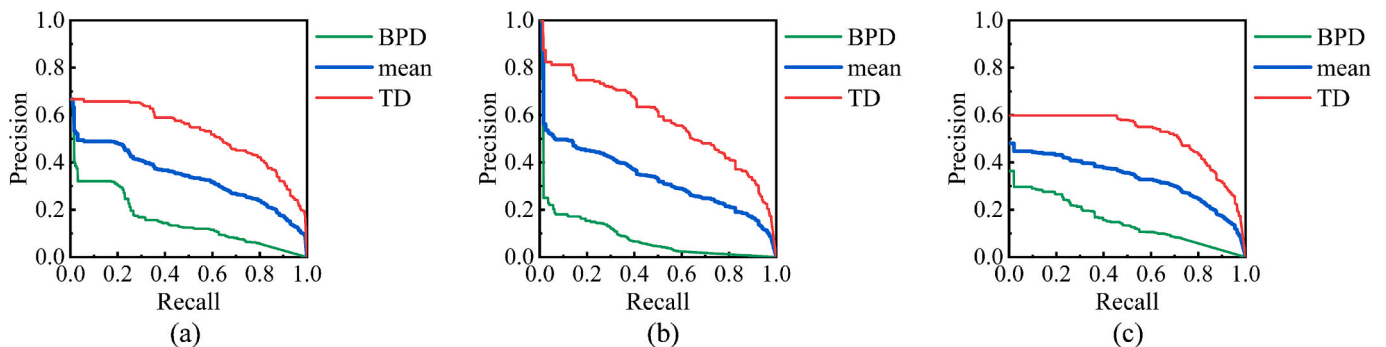


Fig. 16. Precision-Recall (PR) curve comparison of three algorithm models: (a) YOLOv8-OBb, (b) YOLOv10-OBb, and (c) YOLO11-OBb for BPD and TD type dislocations.

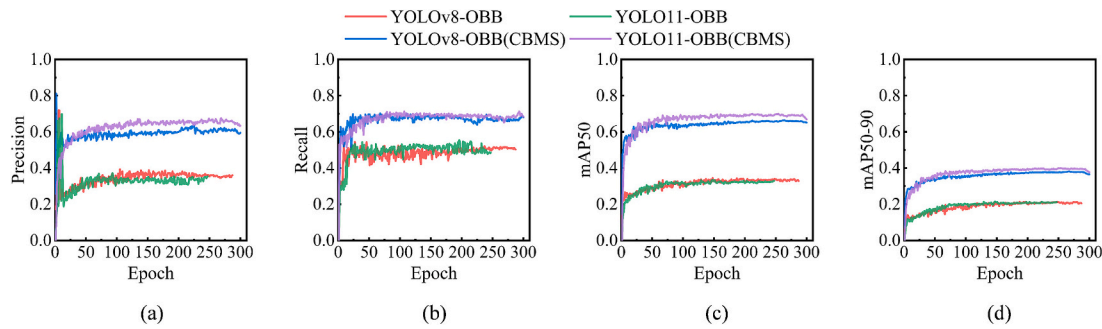


Fig. 17. Performance comparison of YOLOv8-OBb and YOLO11-OBb models with class-balanced multi-sampling (CBMS) strategy: (a) Precision; (b) recall; (c) mAP50; (d) mAP50-90.

particularly critical, as in industrial quality inspection, missed detections are often more severe than false positives. Overall, CBMS's data preprocessing techniques play a crucial role in improving detection performance.

3.3. The detection performance of YOLO11-OBb (CBMS) for TD and BPD

Based on the results of the above experiments, YOLO11-OBb combined with CBMS demonstrated the best performance in 4H-SiC substrate dislocation defect detection. The model detection results for the two types of defects in the substrate are shown in Fig. 18.

From the results in Fig. 18, it can be seen that YOLO11-OBb combined with the CBMS model performs well in the 4H-SiC substrate dislocation defect detection task, especially in predicting the TD category. The precision-recall curve (Fig. 18(a)) shows that both precision and recall for the TD category are at a high level, while the BPDs category performs relatively weakly, particularly in recall, indicating that the model faces significant challenges in detecting BPDs. The confusion matrix (Fig. 18(b)) further confirms this, showing that the correct prediction rate for the TD category has a significantly higher value compared to the BPD category, which indicates that the model has poor discrimination ability for the BPD category, often misclassifying it as a background. Despite the overall mAP50 value of 0.700, further improvements are still needed for BPD detection. To further enhance BPD detection performance, it may be necessary to increase the number of BPDs, adjust the training strategy, or optimize the CBMS module to improve the model's ability to recognize the BPD category, thereby improving the overall detection performance.

Fig. 19 shows the training and validation loss curves of the model, along with various performance metrics. The figure compares the performance of the model throughout the entire training process,

displaying both smoothed and raw curves for different types of loss values.

As shown in Fig. 19, with the continuous development of training work, box_loss, cls_loss, and dfl_loss all decreased rapidly and then reached a stable state. This phenomenon shows that model optimization has made great progress. The curves used to verify these losses were similar to those of the training set. This similarity shows that the model has a good generalization ability when dealing with validation data.

After continuous performance improvement, the performance index finally stabilized, and these results once again proved that the YOLO11-OBb model was reliable and effective for detecting the upper error defect of the 4H-SiC substrate. These results highlight the advantages of this model in improving the accuracy of defect detection and generalization to provide strong support for the application of this model in industrial scenarios.

4. Conclusions

This study introduces a novel approach for identifying dislocation defects in 4H-SiC substrates by employing PL imaging techniques alongside the YOLO object detection algorithm. Due to the fact that dislocation defects in PL images often have low contrast and unclear boundaries, accurate detection is hindered. To overcome these challenges, this study incorporated image enhancement techniques to improve image quality, facilitate easier identification of defects, and utilize the YOLO11-OBb (CBMS) model for defect identification and localization.

The experimental results show that the YOLO11-OBb (CBMS) model outperforms the other models in all evaluation metrics. At the same time, the category distribution in the dataset is improved, so the ability of the model to detect defects in a few categories (e.g., BPD) is also improved. The model has effectively converged and relatively stabilized

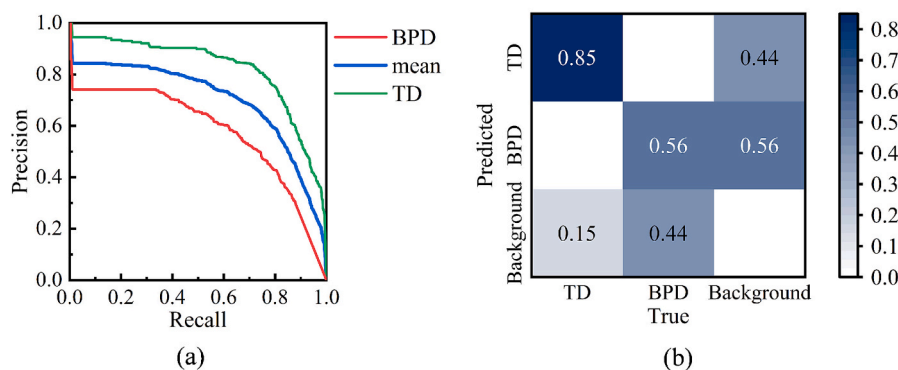
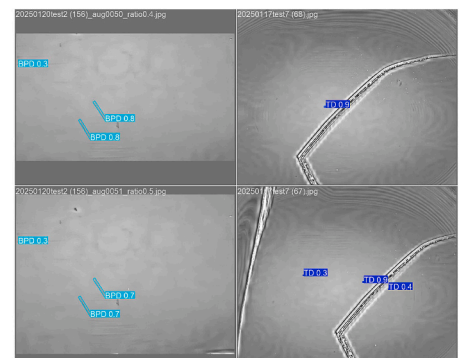


Fig. 18. (a) Precision-recall curve of YOLO11-OBb (CBMS). (b) Confusion matrix of YOLO11-OBb (CBMS). (c) Demonstration of dislocation defect detection using YOLO11-OBb (CBMS).



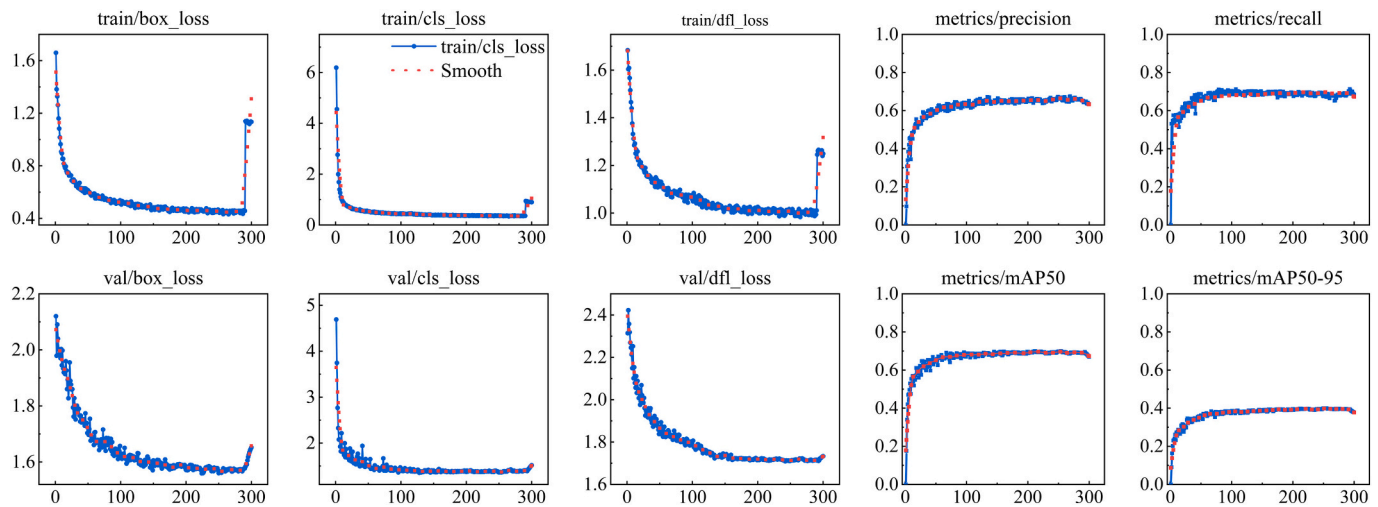


Fig. 19. Results of YOLO11-OB (CBMS).

during the training process, proving that the YOLO11-OB (CBMS) model combined with image enhancement techniques and data pre-processing techniques can effectively and reliably detect dislocation defects in 4H-SiC substrates.

This study illustrates the significant capabilities inherent in the integration of deep learning methodologies with image enhancement techniques, thereby presenting an innovative approach for the automated identification of dislocation defects in 4H-SiC substrates. With additional refinement of the detection model and its adaptation to large-scale production settings, this approach can provide essential assistance in the quality control of 4H-SiC substrates and improve the device performance in future applications.

CRedit authorship contribution statement

Lu-lu Yu: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Yi-Hang Qiu:** Software, Methodology, Formal analysis, Data curation. **Heng Wang:** Writing – review & editing, Visualization, Validation. **Zhen-zhong Wang:** Writing – review & editing, Supervision, Conceptualization. **Zhi-li Hu:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this study.

Data availability

Data supporting the findings of this study are available in the SiC-Dislocation_Detection_DB dataset, which can be accessed through a public repository hosted on the Open Science Framework (OSF) at https://osf.io/5rvfw/?view_only=c3b3aa60c3bb4f60b55faf5fd2c39e07. The dataset includes processed photoluminescence (PL) images of silicon carbide (SiC) substrates, annotated with defect types such as basal plane dislocations (BPDs) and threading dislocations (TDs), along with the corresponding defect detection results utilizing deep learning-based algorithms. The dataset is available for public use under the terms and conditions of the

Open Science Framework platform. Access to the dataset could be requested through the official channels of the repository.

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